

Frequency Down Sampling

1) Objectives of Frequency Down Sampling

- Reduce frequency resolution by binning or decimating frequency bins while preserving spectral characteristics
- Decrease the number of frequency points in Power Spectral Density (PSD) data
- Maintain the original frequency bandwidth (range) during down sampling
- Keep PSD values non-negative and within reasonable bounds

2) How It works

Unlike **time-domain down sampling** (which reduces sampling rate), **frequency-domain down sampling** reduces the number of frequency points in an existing PSD. The process:

1. Takes the current frequency grid (e.g., 0, 0.5, 1.0, 1.5... Hz)
2. Creates a **new, coarser frequency grid** based on target resolution (e.g., 0, 1, 2... Hz)
3. Maps each new frequency point to original PSD values using:
 - a. **Interpolation** (linear) – default method
 - b. **Bin averaging** – weighted average of neighboring bins
 - c. **Decimation** – nearest neighbor selection
4. Preserves the exact frequency range (e.g., 0–100 Hz) while reducing density of points



